

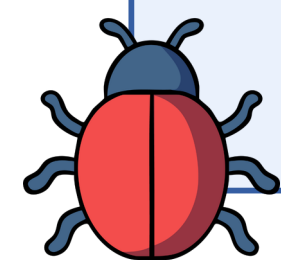
Git Issue: django-12965



Observed: Generates a self-referencing SELECT subquery, causing a performance regression.



Expected: A full delete should execute as a single DELETE query without a SELECT subquery.



Buggy SQL: DELETE..WHERE id IN (SELECT..)

After Patch: DELETE FROM table

State of The Art: SpecRover

Intended Behavior

Return True when **at most one alias** is referenced (i.e. $\sum(\text{self.query.alias_refcount}[t] > 0 \text{ for } t \text{ in self.query.alias_map}) \leq 1$).

Selected Patch

```
def single_alias(self):  
- return sum(self.query.alias_refcount[t] >  
0 for t in self.query.alias_map) == 1  
+ return sum(self.query.alias_refcount[t] >  
0 for t in self.query.alias_map) ≤ 1
```

Reviewer

Approved!

Evaluator

UNRESOLVED

Tests F2P: 0/1 P2P: 51/52

❗ Repair intent lacks an executable specification. ❗ A plausible patch may leave the reported bug unfixed.

Our Approach: ContractFix

① Natural Language Repair Contract (C_{NL})

Precondition: SQLDeleteCompiler.as_sql() compiles a simple Model.objects.all().delete() query **without joins or multiple tables**.
Postcondition: as_sql() returns a **direct DELETE** without a WHERE pk IN (SELECT ...) subquery.

② Executable Contract (C_E)

```
...  
sql, params = compiler.as_sql()  
...  
assert "WHERE" not in sql.upper()  
assert "SELECT" not in sql.upper()
```

✓ Runs in repository context ✓ Violated by buggy code
✓ Reaches issue-relevant behavior ✓ Review: Conforms to C_{NL}

③ C_{NL} Guided Patch Candidate A

```
- return sum(self.query.alias_refcount[t] > 0..  
+ return len(self.query.alias_map) == 1
```

④ Patch Validation

	PA	PB
Patch applicable	✓	✓
Syntax valid	✓	✓
No new test failures	✓	✓
Satisfies Repair C_E	✗	✓

⑤ C_E Validation Report

Violated: assert "WHERE" not in sql.upper()
Observed: DELETE FROM "auth_user" WHERE..
Feedback: Model.objects.all().delete() must generate a direct DELETE without a self-referencing subquery.

⑥ C_E Refined Patch B

```
def single_alias(self):  
+ self.query.get_initial_alias()  
return sum(self.query.alias_...
```

Evaluator

RESOLVED Tests F2P: 1/1 P2P: 52/52

✓ Repair intent becomes executable
✓ Intended behavioral delta is validated
✓ Executable evidence justifies the repair